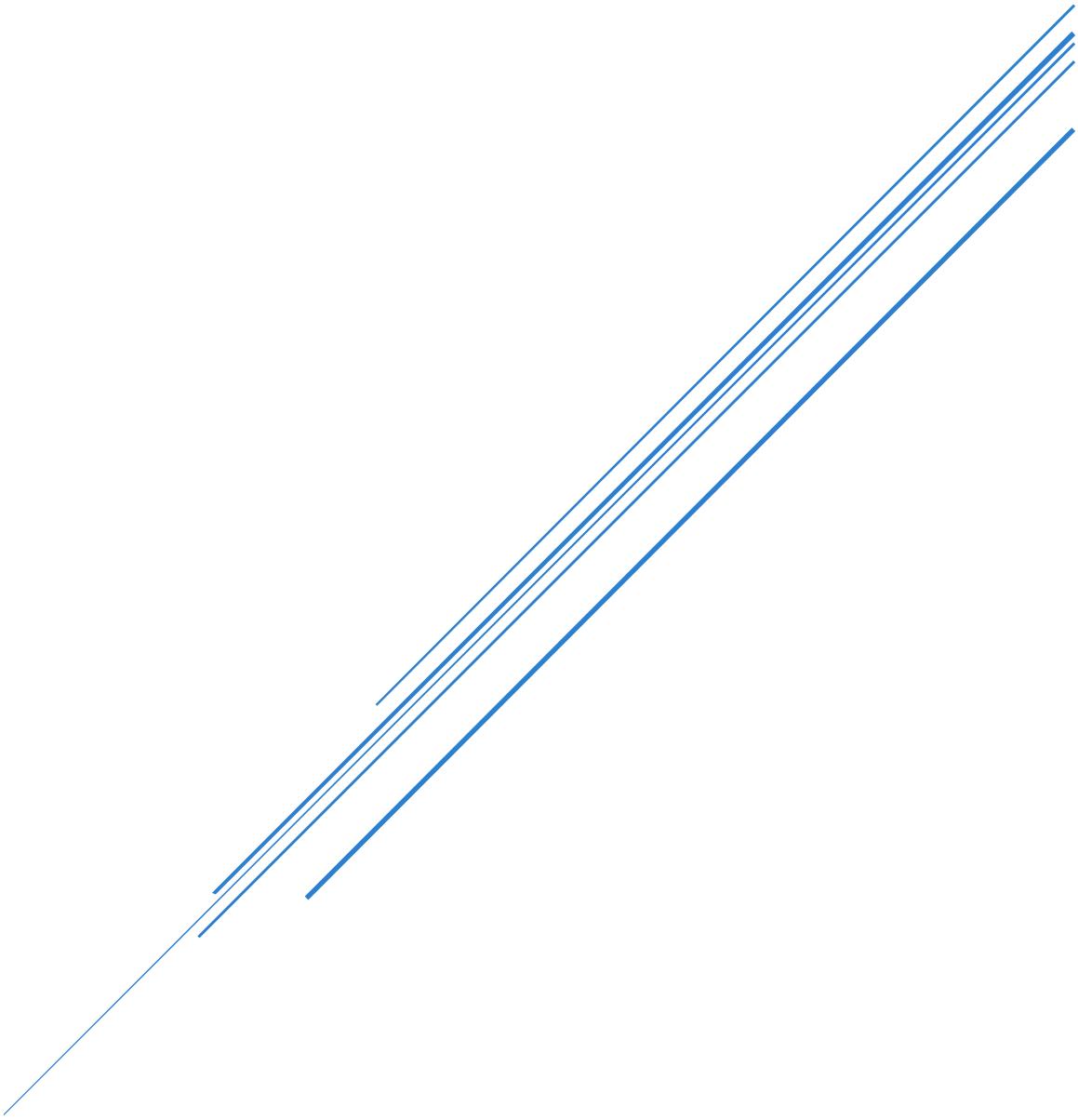
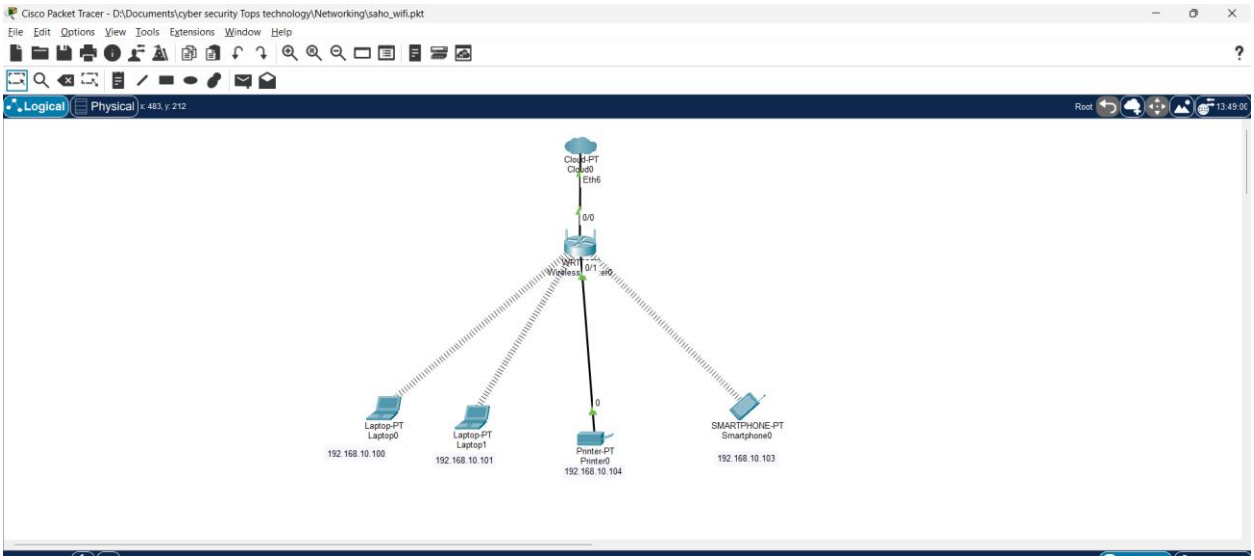
BUILD A SAHO NETWORK

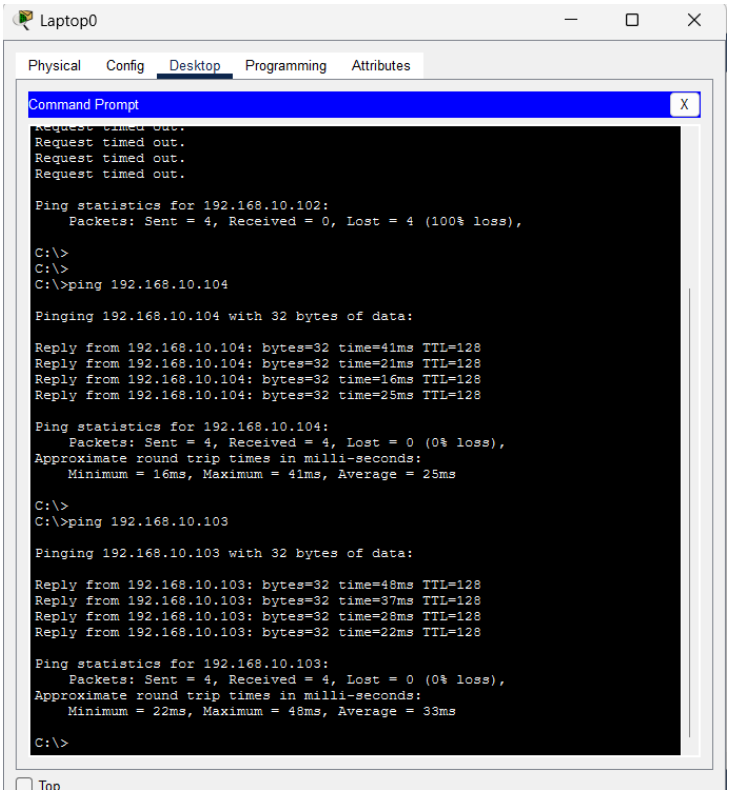
Assignment - 1



1. **Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).**

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2. **List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.**
   Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

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Hardware Components Required for SOHO Network

No.	Hardware Component	Brand / Model
1	Broadband Internet	Airtel Xstream / JioFiber
2	Modem / ONT	ISP-provided ONT
3	Wi-Fi Router	TP-Link Archer C6
4	Laptops (2)	HP / Lenovo
5	Smart TV (1)	Samsung Smart TV
6	Printer (1)	HP LaserJet
7	Ethernet Cable	Cat6 LAN Cable
8	Network Switch (Optional)	TP-Link TL-SG1005D
9	Wi-Fi Extender (Optional)	TP-Link RE315
10	UPS (Optional)	APC UPS

3. **Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.**

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1. Connect Router: Connect the broadband modem to the router's WAN/Internet port.

2. Open Router Settings: Connect your laptop to the router and open a browser. Enter 192.168.0.1 or 192.168.1.1.

3. Login: Enter the default admin username and password provided by the router manufacturer.

4. Set SSID: Go to Wireless Settings and set a unique Wi-Fi name:

- **SSID: SOHO_WIFI**

5. Enable WPA2 Security:

- **Security Mode: WPA2-PSK**
- **Encryption: AES**
- **Password: Soho@12345**

6. Change Admin Password: Go to Administration/System Tools and change the default admin password to a strong, unique password.

7. Save Settings: Click Save/Apply and restart the router if required.

8. Connect Devices: Connect both laptops and Smart TV using the new Wi-Fi name and password.

4. **Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.**

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Network Details:

- **Network: 192.168.10.0/24**
- **Subnet Mask: 255.255.255.0**
- **Default Gateway: 192.168.10.1**
- **DNS Server: 8.8.8.8**

IP Address Allocation Table

Device	MAC Address (Sample)	IP Address	Method
Wi-Fi Router	00:1A:2B:3C:4D:01	192.168.10.1	Static
Laptop 1	00:1A:2B:3C:4D:02	192.168.10.100	DHCP
Laptop 2	00:1A:2B:3C:4D:03	192.168.10.101	DHCP
Smart TV	00:1A:2B:3C:4D:04	192.168.10.10	Static
Printer	00:1A:2B:3C:4D:05	192.168.10.20	Static

Configuration Steps

1. **Router: Set LAN IP to 192.168.10.1.**
2. **DHCP: Enable DHCP and set the range from 192.168.10.100 to 192.168.10.150.**
3. **Laptops: Select DHCP in IP Configuration to obtain automatic IP addresses.**
4. **Smart TV: Manually assign IP 192.168.10.10.**
5. **Printer: Manually assign IP 192.168.10.20.**
6. **Gateway: Set 192.168.10.1 on both static devices.**
7. **DNS: Set 8.8.8.8.**

5. **Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.**

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1. Incorrect Wi-Fi Password

- **Cause: Wrong Wi-Fi password.**
- **Solution: Check the password and reconnect to the correct SSID (SOHO_WIFI).**

2. Incorrect IP Configuration

- **Cause: Laptop is not receiving a valid IP address.**
- **Solution: Open IP Configuration and select DHCP to obtain an automatic IP address.**

3. Router Problem

- **Cause: Router is switched off or not working properly.**
- **Solution: Check the power and restart the router.**

4. Weak Wi-Fi Signal

- **Cause: Laptop is too far from the router.**
- **Solution: Move closer to the router or use a Wi-Fi extender.**

5. DNS Problem

- **Cause: DNS server is not resolving website names.**
- **Solution: Set DNS to 8.8.8.8 (Google DNS).**